

KENNEDY MUNENE

+254 795 396 241 | kennedydsm@gmail.com | [LinkedIn](#) | [GitHub](#) | Open to Relocate

SUMMARY

Backend Developer and Data Scientist with hands-on experience shipping production machine learning systems, FastAPI/Flask backends, and full-stack web platforms. Built and deployed an insurance fraud detection system (ROC-AUC 0.78), a multi-region professional services platform, and an offline-first task management app - combining ML modeling, database design, and API engineering. Skilled in Python, SQL, React/TypeScript, and Agile delivery from prototype to production.

EDUCATION

Bachelor of Science in Software Engineering, Murang'a University of Technology Graduation: 2026
DataCamp Certified Associate Data Scientist – Sep 2024

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, HTML/CSS
Backend & APIs: FastAPI, Flask, Django, REST APIs, SQLAlchemy, Alembic, JWT Auth, Microservices
ML / Data Science: TensorFlow, PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, SHAP, Model Deployment
Frontend: React, TypeScript, Zustand, Jinja2, Recharts, Responsive UI/UX
Databases & Infra: PostgreSQL, MySQL, SQLite, Supabase, Docker, Git, CI/CD, Render, Vercel

PROJECTS

- FraudGuard AI – Insurance Fraud Detection System** | [Live Demo](#) Apr 2026
- Built an end-to-end fraud detection system on 15,420 motor insurance claims; **XGBoost model achieves ROC-AUC 0.781** with a business-calibrated threshold delivering a **72.5% fraud detection rate** and an estimated **\$294K in savings per 2,300 claims**.
 - Engineered a **FastAPI** backend with **PostgreSQL** (async SQLAlchemy + Alembic), persisting every prediction with a full audit trail for regulatory compliance.
 - Designed **76 domain-specific features** (fault attribution flags, agent interaction patterns, claim-timing signals); top SHAP drivers align with known motor-fraud patterns.
 - Deployed a **React + TypeScript** frontend with real-time risk scoring, SHAP-based explanations, and an analytics dashboard of fraud trends.
- Dr. Jacob Advisory – Professional Services & Portfolio Platform** | [Live Demo](#) Feb 2026 – Mar 2026
- Built a full-stack platform for an international clientele across Africa, Asia, Europe, and the Americas, featuring real-time testimonials, a blog CMS, and visitor analytics.
 - Engineered a **FastAPI** backend with a four-layer architecture (router, service, repository, database), **JWT authentication**, and five-layer spam prevention including server-side hCaptcha verification.
 - Designed a **PostgreSQL** database on **Supabase** with Row Level Security, real-time WebSocket subscriptions, and Alembic-managed migrations across seven production tables.
 - Built a **React/TypeScript** frontend (Zustand, TipTap rich-text editor, real-time listeners, full dark/light mode) plus a private admin dashboard with Recharts analytics, blog management, and testimonial moderation.
- GetItDone – Full-Stack Task Management Platform** | [Live Demo](#) Nov 2025 – Dec 2025
- Developed a production-ready task management app with a **FastAPI** backend, deployed on Render with **99%+ uptime**; architected an offline-first system with local caching and automatic conflict resolution.
 - Implemented secure authentication for guest and registered accounts with encrypted data handling, plus a real-time push notification system via **Firebase Cloud Messaging** for cross-device reminders.
- FlashHome AI – Real Estate Pricing & Market Insights Platform** | [Live Demo](#) Aug 2025
- Built an AI-powered real estate platform for instant house price predictions; developed the **Flask** backend with integrated ML models, deployed via **Docker** on Render, and designed an Admin Dashboard to track inputs and trends.

PROFESSIONAL EXPERIENCE

IT Intern, Murang'a Level 5 Hospital Apr 2024 – Jul 2024

- Assisted with backend system maintenance, database administration, and troubleshooting across hospital information systems.